

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test
COURSE NAME: Cloud Computing and Big Data Analytics
COURSE CODE: CSA15

89
100

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping
COURSE NAME: Cloud Computing and Big Data Analytics
COURSE CODE: CSA15

90
100

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct

I S. Subashini (192511296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Subashini

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Total Marks Awarded:

Signature of the Faculty

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy. Include front-end, back-end, client devices, data centers, and distributed servers.

1. Concept Map 1: Cloud Computing Architecture: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

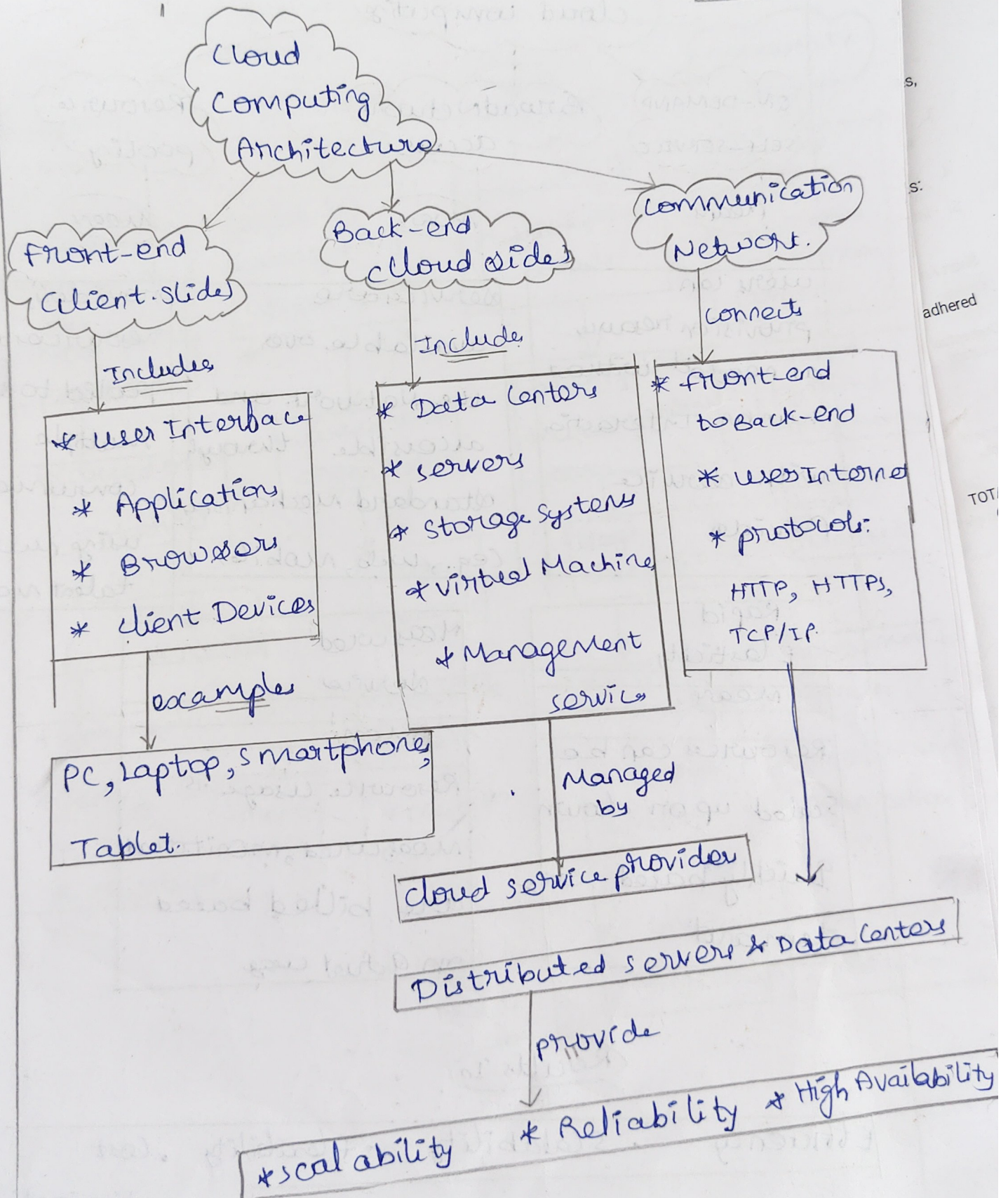
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

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Reg. NO: 19251129
COT Assessment 2

CONCEPT MAP 1: CLOUD COMPUTING ARCHITECTURE CSA 1521

cloud computing



2)

CONCEPT MAP2: CHARACTERISTICS OF CLOUD COMPUTING

characteristics of cloud computing

ON-DEMAND
SELF-SERVICE

Broad Network
access

Resource
pooling

means

means

means

users can provision resources as needed without human interaction with service provider.

services are available over the Network and accessible through standard mechanisms (eg., web, mobile).

providers, resources are pooled to serve multiple consumers using multi-tenant model

Rapid
Elasticity

means

Resources can be scaled up or down quickly based on demand

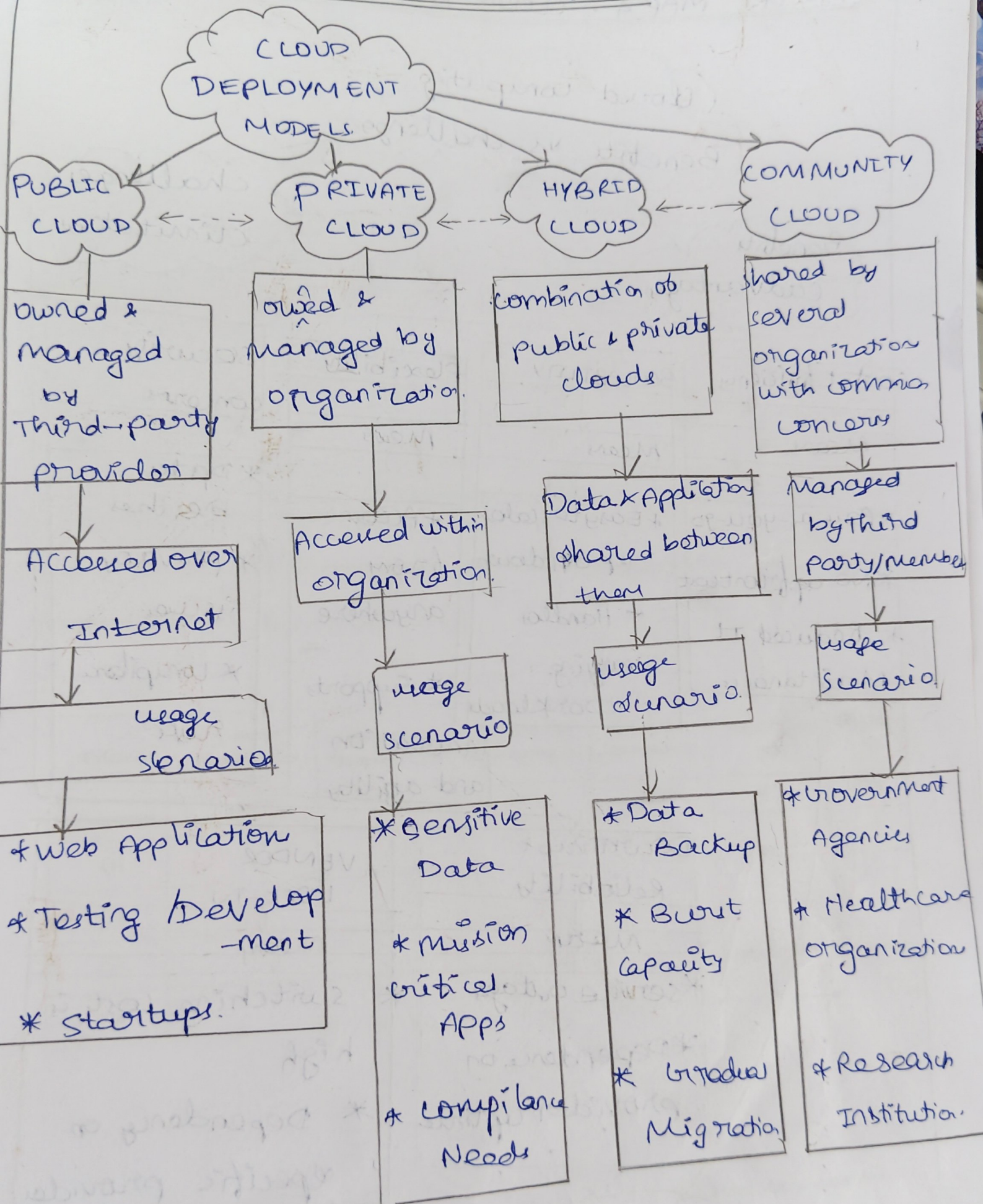
Measured
service

means

Resource usage is measured, monitored and billed based on actual usage

Results In

CONCEPT MAP 3: DEPLOYMENT MODELS



4) CONCEPT MAP 4: CLOUD BENEFITS VS CHALLENGES

